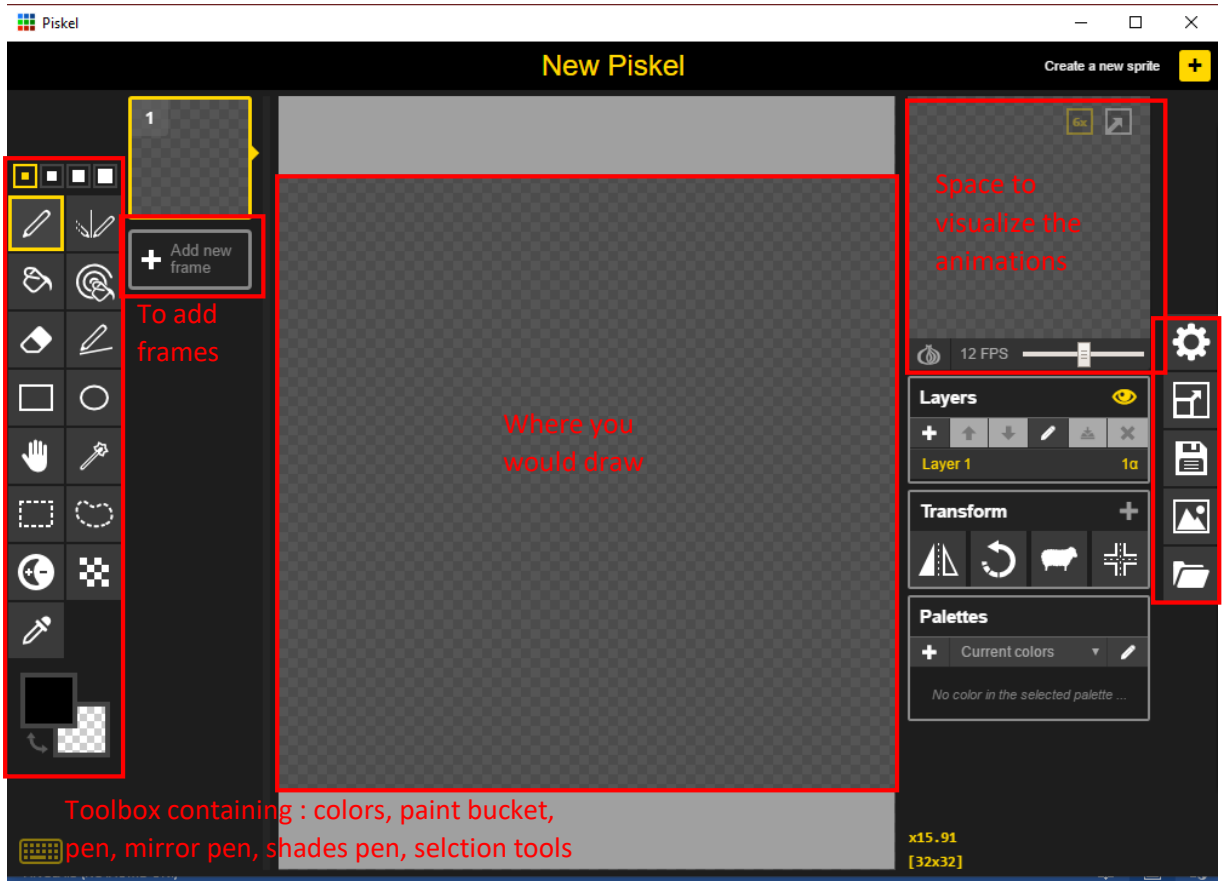
PW 1 - Sprites

Sprite Creation and Visualization :

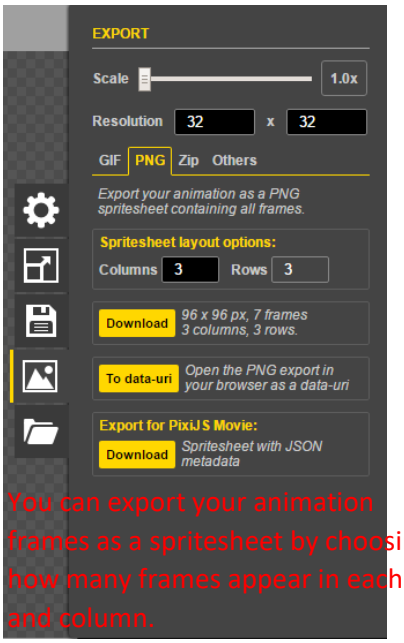
Goal: Use « Piskel » to draw a sequence of frames of your own sprite charcater. Export the sprite into a .png file.

Use « Canvas » and Javascript, display your sprite as an animation in an HTML page.

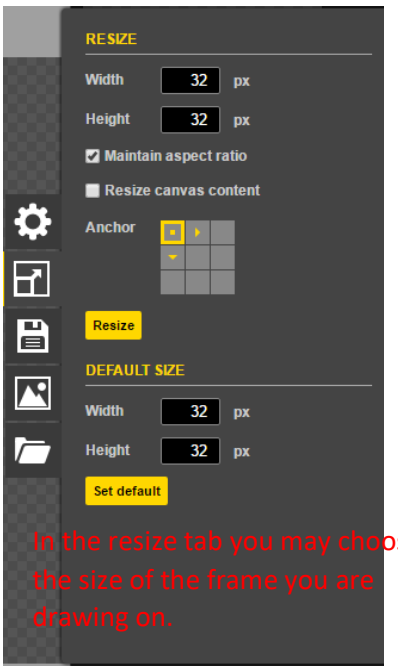
Step 1: Drawing multiple frames for one sprite



Step 2: Export your frames into one .png spritesheet



You can export your animation frames as a spritesheet by choosing how many frames appear in each row and column.



In the resize tab you may choose the size of the frame you are drawing on.

Step 3: Visualize using HTML canvas and Javascript

a. Start by displaying the whole spritesheet on the canvas;

```
image = new Image();
image.src = "spritesheet_path.png";
image.onload = function () {
    ctx.drawImage(image, x, y);

};
```

b. Now, display only the first frame of the sprite on the canvas;

```
drawImage(image, i, j, width, height, x, y, width, height);
where: (i, j) coordinates of frame on spritesheet,
       (x, y) coordinates of frame on the canvas,
       (width, height) dimensions of the sprite (frame).
```

c. Write the loop that displays all the frames one by one. What do you notice when refreshing the browser?

When the loop executes, all frames are drawn immediately, one after another, on the same canvas. Since the canvas retains previously drawn content, the frames overlap.

Correct it by clearing the canvas. `ctx.clearRect();`

Now, only the last frame remains visible. The loop executes only once and does not produce an animation.

d. Fix the issue using the appropriate function.

We need a function that redraws the screen on every frame. The game loop must run continuously, indefinitely, as long as the game is running. User either:

`requestAnimationFrame(gameloop);` or:

`setInterval(gameloop, delay);`